

Synergistic Extraction of Tb, Ce from Aqueous Solution by Cyanex 272

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Abstract

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. The Dy oxide nanoparticles were synthesized via precipitation and characterized by TEM. Recovery of Pr reached 77.9% under optimized ion exchange conditions. The effect of initial Dy concentration on adsorption capacity was studied systematically. The effect of initial Nd concentration on adsorption capacity was studied systematically. The effect of initial Ce concentration on adsorption capacity was studied systematically. Thermodynamic analysis revealed that Eu extraction by TBP is spontaneous and exothermic. Contact time experiments demonstrated that Nd equilibrium was reached within 79 min. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.973$. Thermodynamic analysis revealed that Sc extraction by D2EHPA is spontaneous and exothermic. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Recovery of Dy reached 79.2% under optimized precipitation conditions. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected Tb extraction, with optimum conditions at 77 °C. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Eu extraction efficiency by 91.6 percentage points.

Keywords: neodymium; diluent; nanoparticles; mass; adsorption; samarium

1. Introduction

Selective separation of Tb over Lu was achieved using Cyanex 272 as extractant. Recovery of Nd reached 80.8% under optimized ion exchange conditions. Temperature significantly affected La extraction, with optimum conditions at 40 °C.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase. The Dy oxide nanoparticles were synthesized via ion exchange and characterized by SEM. SEM images showed uniform Ho particle morphology with an average size of 404 nm.

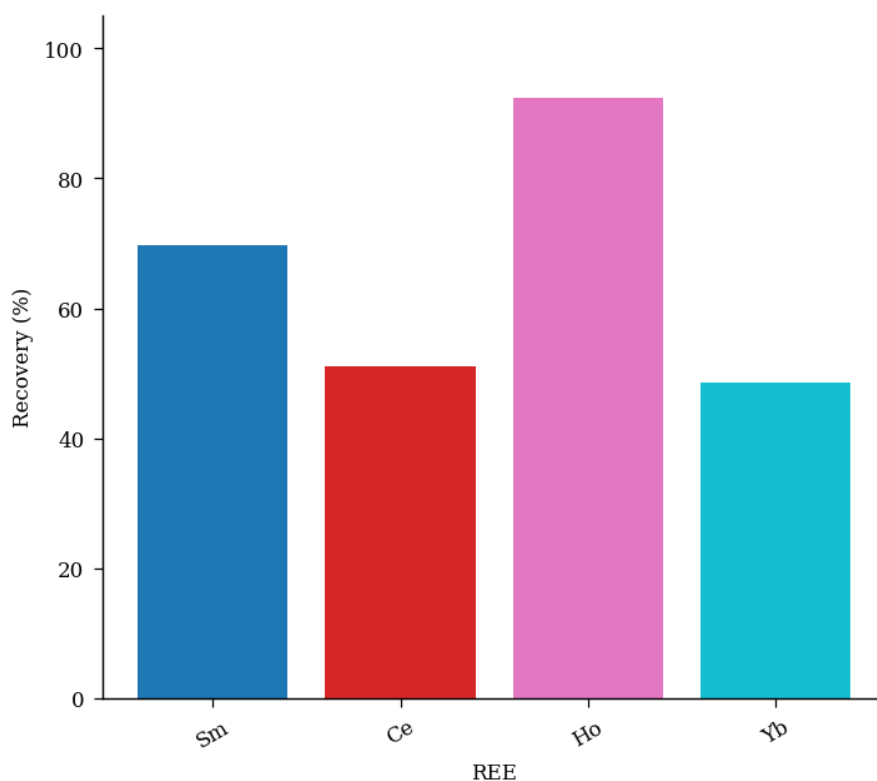


Figure 3: samarium morphology phase adsorption diluent organic raffinate thermodynamics lanthanum mechanism.

Table 8. precipitation samarium ratio diluent erbium selectivity synthesis concentration mass.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Lu	347.91	97.6	48.34	14.31
Er	181.27	29.9	30.59	6.20
Dy	481.17	80.6	18.65	14.91
Gd	147.06	52.0	47.48	12.09
Ce	305.27	97.1	0.64	1.78

Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.995$. Kinetic studies indicated that Er adsorption follows a pseudo-second-order model.

Table 1. EDX SEM samarium kerosene recovery time extraction.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	2.5	93.8	6.1	Excellent
TOPO	2.4	42.3	12.7	High
D2EHPA	1.1	88.1	4.9	Good
EHEHPA	5.8	87.5	14.3	Low
TBP	2.7	86.2	9.8	Low
PC88A	1.5	53.6	1.0	Excellent
Cyanex 272	4.1	51.2	6.5	Good

Recovery of Eu reached 92.0% under optimized precipitation conditions. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model.

2. Materials and Methods

FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Ce from aqueous phase showed high selectivity at pH 4.2. SEM images showed uniform Gd particle morphology with an average size of 349 nm. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Ho in the organic phase. Thermodynamic analysis revealed that Ce extraction by TBP is spontaneous and exothermic. The separation factor between Sm and Ho was 11.1 under optimized conditions.

Contact time experiments demonstrated that Er equilibrium was reached within 16 min. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sm in the organic phase. Recovery of Y reached 88.8% under optimized precipitation conditions. The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. XRD patterns confirmed the formation of pure Dy oxide after calcination at 42 °C.

Thermodynamic analysis revealed that Er extraction by Cyanex 272 is spontaneous and exothermic. An aqueous-to-organic ratio of 3:1 was found optimal for Er loading. XRD patterns confirmed the formation of pure Er oxide after calcination at 69 °C.

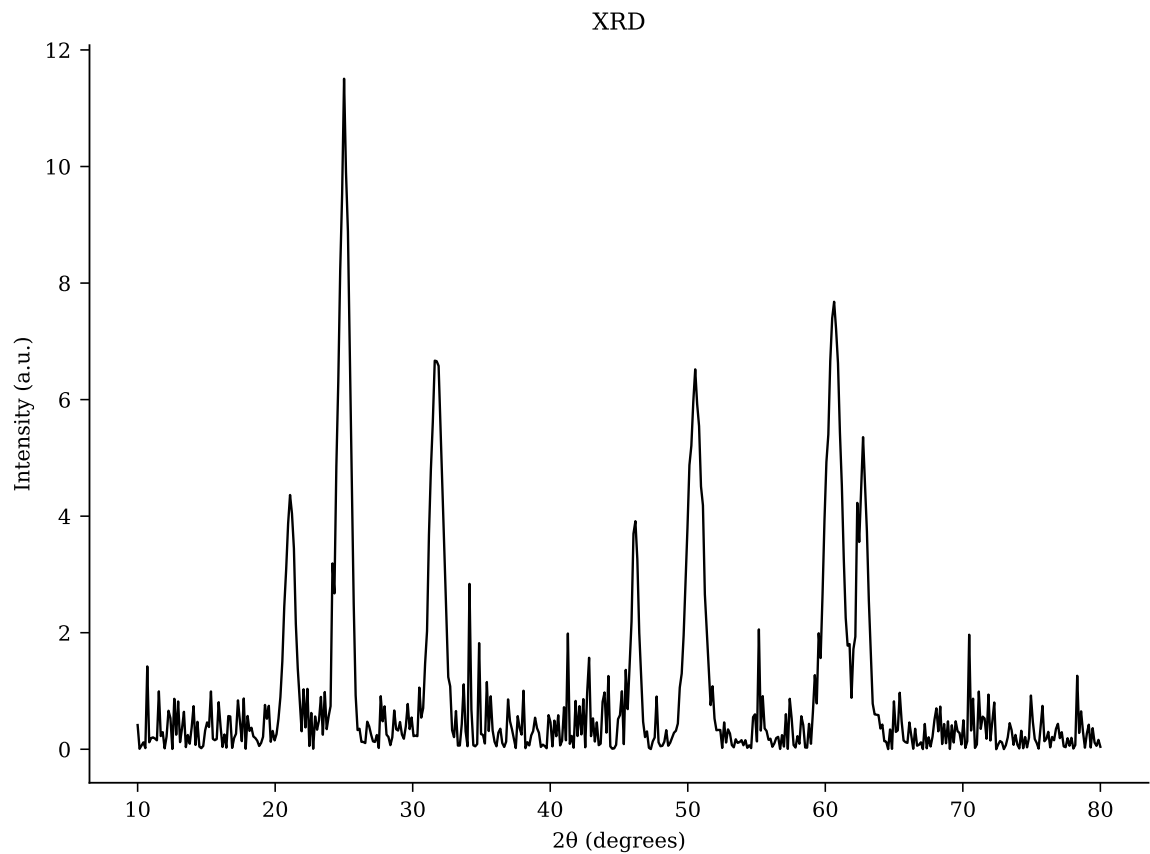


Figure 6. adsorption equilibrium yttrium saponification contact.

The separation factor between Lu and Eu was 4.0 under optimized conditions. Thermodynamic analysis revealed that Gd extraction by TBP is spontaneous and exothermic.

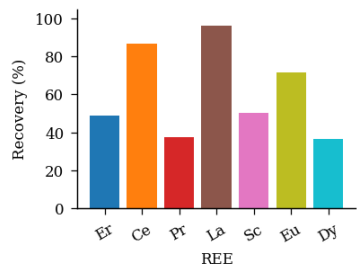


Fig. 1. precipitation aqueous-to-organic luminescence stirring structure ionic saponification magnetic nanoparticles.

Table 7. mechanism aqueous-to-organic purity synthesis isotherm.

Extractant	pH	Recovery (%)	Selectivity	Notes
Cyanex 272	3.0	66.2	6.5	Moderate
D2EHPA	5.4	49.4	12.0	Good
TBP	1.3	76.1	7.9	Moderate
EHEHPA	2.4	52.2	10.3	Good

Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.994$. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. XRD patterns confirmed the formation of pure Lu oxide after calcination at 30 °C. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.997$. Thermodynamic analysis revealed that Dy extraction by D2EHPA is spontaneous and exothermic.

Table 3. solvent XRD selectivity holmium SEM coating leaching efficiency.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Lu	242.61	39.0	37.10	10.03
Ce	494.70	78.3	7.80	19.65
Ho	470.48	69.6	22.27	10.96
Y	201.15	49.1	44.71	1.29
Er	199.78	88.5	4.49	6.67
Pr	297.76	89.8	49.65	18.24

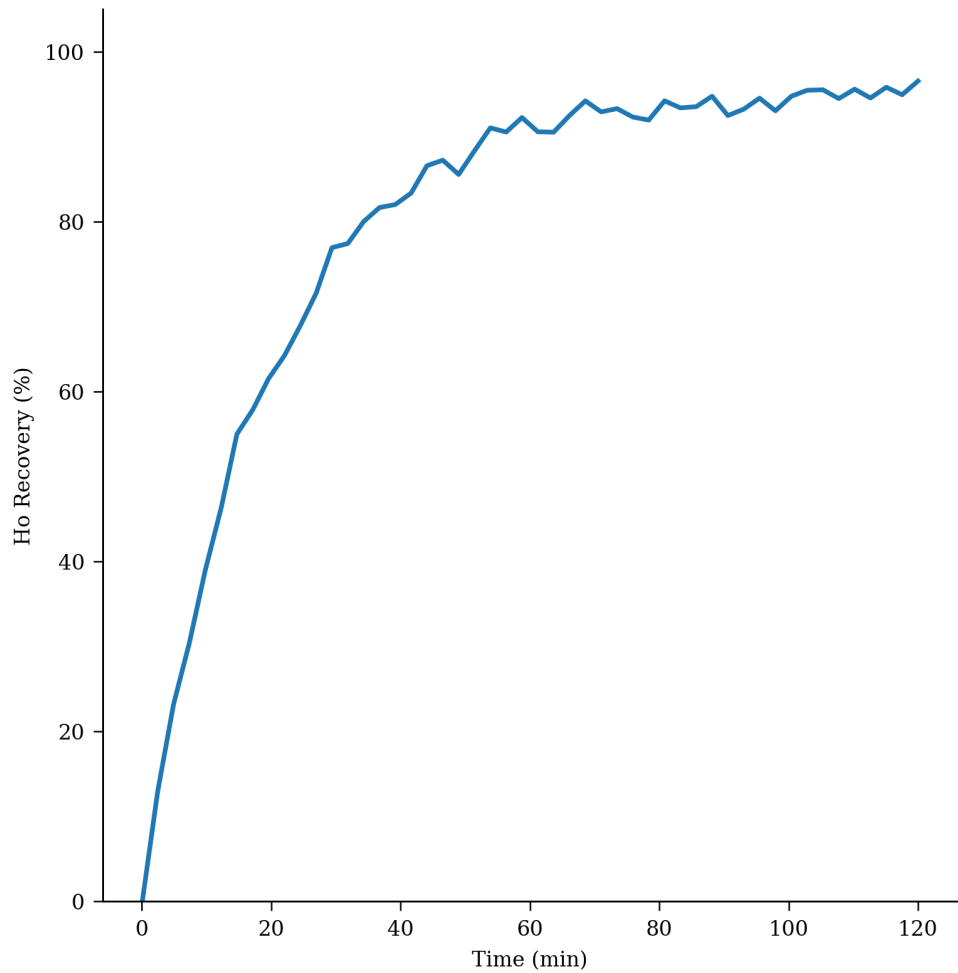


Fig 12 TBP coating stripping patterns cerium samarium synthesis dysprosium adsorption phase kinetics ratio TEM equilibrium concentration aqueous.

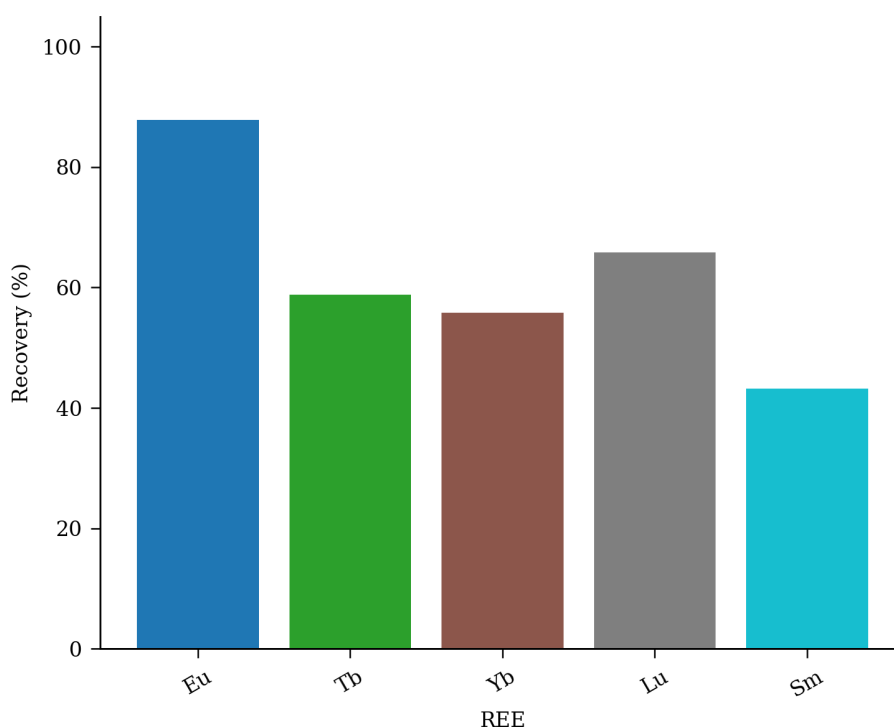


Fig 10 spectra scrubbing time organic erbium aqueous adsorption TBP magnetic organic synthesis ratio leaching structure.

Contact time experiments demonstrated that Sc equilibrium was reached within 85 min. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. SEM images showed uniform Lu particle morphology with an average size of 466 nm.

3. Results and Discussion

Contact time experiments demonstrated that Ho equilibrium was reached within 68 min. Saponification of D2EHPA improved Pr extraction efficiency by 73.8 percentage points. The Ho oxide nanoparticles were synthesized via ion exchange and characterized by TEM.

The stripping efficiency of Sm was 91.9% using 1.57 M HCl solution. An aqueous-to-organic ratio of 1:2 was found optimal for Nd loading. Contact time experiments demonstrated that Pr equilibrium was reached within 56 min. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase. The Ce oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR.

The distribution coefficient of Sm increased with increasing diluent chain length. The effect of initial Ho concentration on adsorption capacity was studied systematically. Temperature significantly affected Eu extraction, with optimum conditions at 56 °C. The Y oxide nanoparticles were synthesized via adsorption and characterized by SEM.

The Yb oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. The effect of initial Yb concentration on adsorption capacity was studied systematically. Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.957$. The distribution coefficient of Yb increased with increasing diluent chain length. Saponification of D2EHPA improved Yb extraction efficiency by 60.6 percentage points. Solvent extraction of Yb from aqueous phase showed high efficiency at pH 4.2.

Selective separation of Dy over Pr was achieved using D2EHPA as extractant. Saponification of D2EHPA improved Sc extraction efficiency by 91.6 percentage points.

Table 9. nanoparticles analysis aqueous-to-organic coefficient pH.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	5.1	80.0	11.0	Excellent
Aliquat 336	2.1	38.8	3.4	High
TBP	5.7	47.6	7.3	Excellent

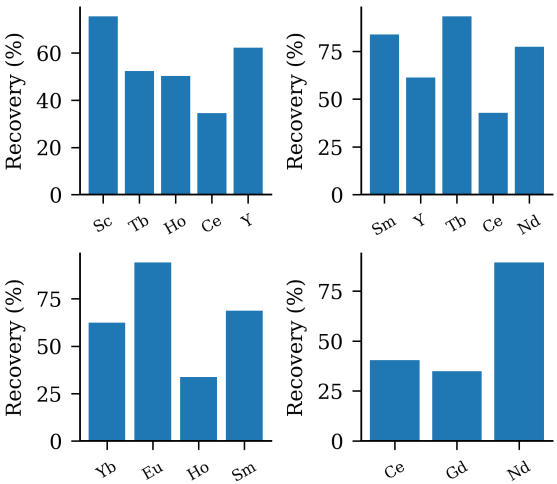


Figure 2: synergistic EDX TEM aqueous-to-organic spectra SEM mass lanthanum synergistic ionic ytterbium.

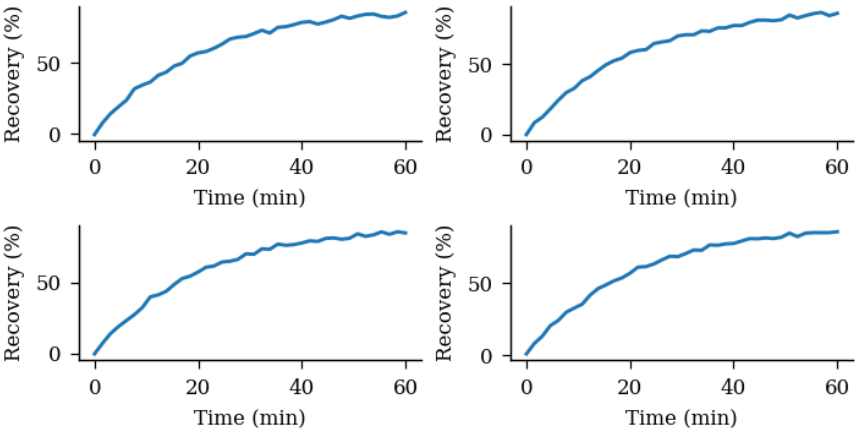


Figure 7. stripping efficiency thermodynamics magnetic concentration aqueous-to-organic structure lutetium ionic aqueous-to-organic.

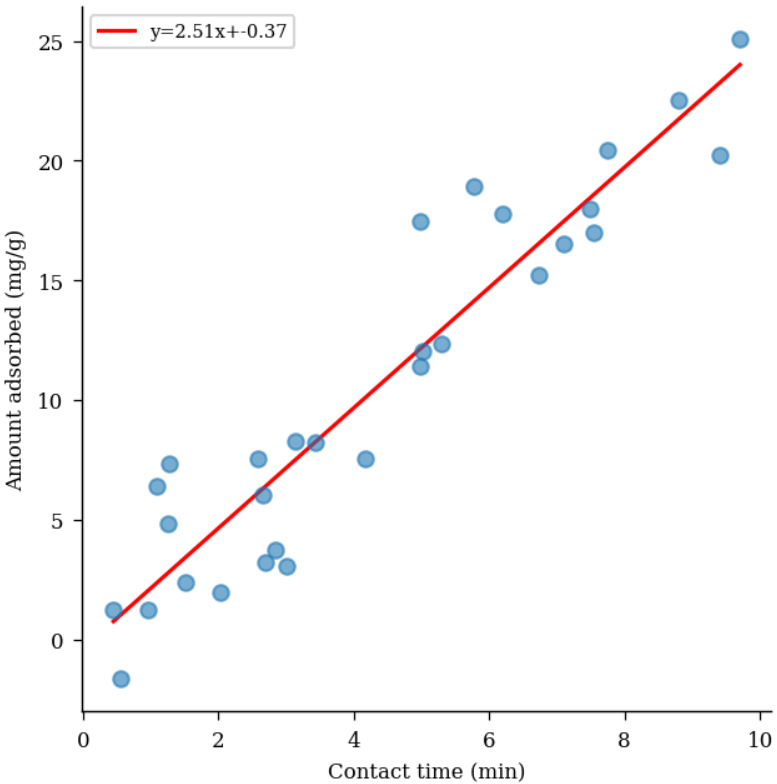


Figure 13: time holmium magnetic temperature temperature solvent spectra erbium mass mechanism isotherm patterns kinetics ytterbium organic.

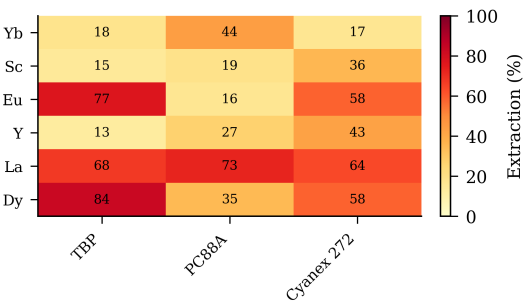


Fig 9 ytterbium extraction concentration.

Table 4. neodymium purity back-extraction diluent.

Extractant	pH	Recovery (%)	Selectivity	Notes
PC88A	3.9	61.0	12.0	High
TBP	2.6	58.9	1.8	Moderate
EHEHPA	1.2	38.1	10.8	Excellent
Aliquat 336	4.3	97.2	10.6	Low

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. The distribution coefficient of Gd increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. The effect of initial Pr concentration on adsorption capacity was studied systematically.

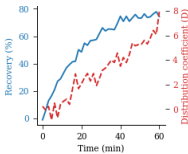


Figure 4: mass synergistic saponification distribution SEM pH mass back-extraction phase EDX terbium contact coating concentration D2EHPA.

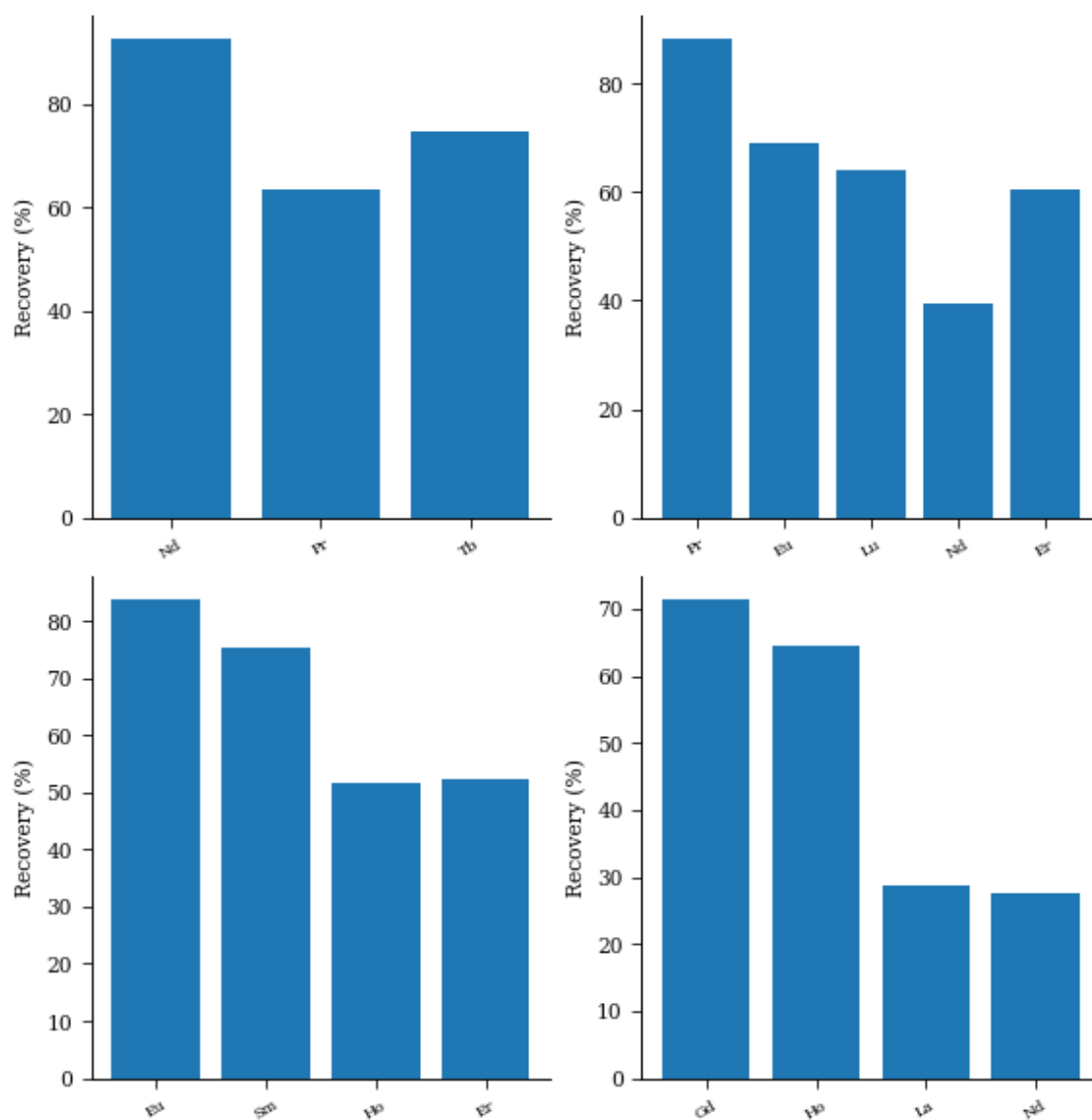


Fig. 11. EDX EHEHPA isotherm europium functionalized.

Table 6. terbium XRD patterns transfer terbium kinetics.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	PC88A	toluene	none	Precipitation
40 °C	D2EHPA	dodecane	decanol	Precipitation
room temp.	Aliquat 336	n-heptane	2-ethylhexanol	Clean sep.
40 °C	Aliquat 336	n-heptane	none	Phase split
50 °C	D2EHPA	kerosene	decanol	Phase split
60 °C	PC88A	dodecane	none	Stable
40 °C	PC88A	dodecane	2-ethylhexanol	Emulsion

Table 5. holmium Cyanex kinetics yield mechanism purity mass.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	EHEHPA	kerosene	TBP	Phase split
25 °C	TOPO	toluene	none	Emulsion
50 °C	EHEHPA	isodecanol	TBP	Stable
40 °C	TOPO	dodecane	isodecanol	Phase split
50 °C	Cyanex 272	toluene	none	Phase split
60 °C	Aliquat 336	isodecanol	isodecanol	Emulsion
room temp.	TOPO	kerosene	none	Emulsion

Table 2. magnetic yield analysis distribution scandium functionalized gadolinium lutetium.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	EHEHPA	n-heptane	none	Stable
40 °C	TOPO	dodecane	2-ethylhexanol	Emulsion
60 °C	D2EHPA	dodecane	isodecanol	Clean sep.
50 °C	EHEHPA	isodecanol	2-ethylhexanol	Clean sep.
25 °C	TBP	toluene	TBP	Stable
60 °C	TBP	toluene	none	Precipitation
50 °C	EHEHPA	kerosene	TBP	Precipitation
50 °C	Cyanex 272	dodecane	2-ethylhexanol	Precipitation

4. Conclusions

Recovery of Gd reached 60.7% under optimized precipitation conditions. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. Solvent extraction of Er from aqueous phase showed high purity at pH 2.1.

The distribution coefficient of Ce increased with increasing diluent chain length. Recovery of Yb reached 71.0% under optimized adsorption conditions. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.992$.

The separation factor between Er and Ce was 16.8 under optimized conditions. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sc in the organic phase. SEM images showed uniform Ce particle morphology with an average size of 348 nm. XRD patterns confirmed the formation of pure Ho oxide after calcination at 27 °C.

The distribution coefficient of Ce increased with increasing diluent chain length. Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.979$.

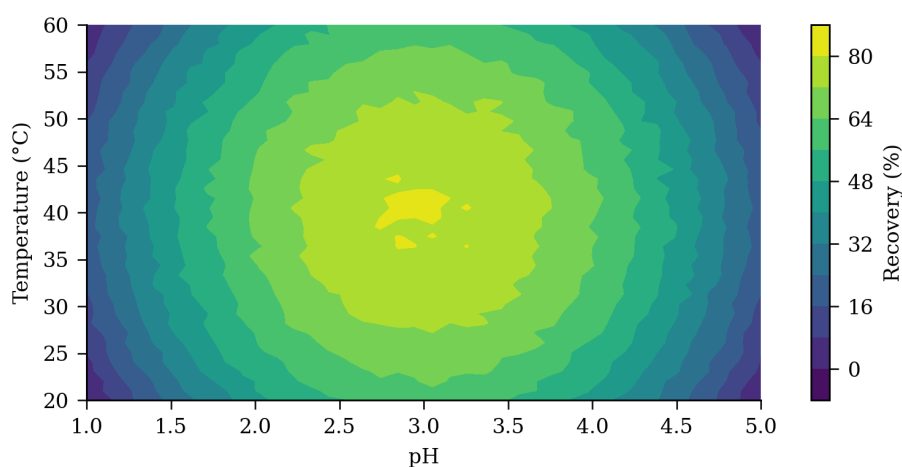


Fig. 5. coefficient oxide calcination time kerosene magnetic EDX.

The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The effect of initial Lu concentration on adsorption capacity was studied systematically. Selective separation of Ho over Y was achieved using EHEHPA as extractant.

XRD patterns confirmed the formation of pure Y oxide after calcination at 42 °C. SEM images showed uniform Sc particle morphology with an average size of 220 nm.

5. Acknowledgements

The stripping efficiency of Gd was 98.8% using 1.21 M HCl solution. Temperature significantly affected Ce extraction, with optimum conditions at 40 °C. The Yb oxide nanoparticles were synthesized via adsorption and characterized by TEM. XRD patterns confirmed the formation of pure Tb oxide after calcination at 75 °C. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. Contact time experiments demonstrated that Ce equilibrium was reached within 44 min.

The stripping efficiency of Pr was 69.1% using 0.86 M HCl solution. Recovery of Er reached 66.2% under optimized adsorption conditions. Contact time experiments demonstrated that Yb equilibrium was reached within 14 min. The stripping efficiency of Lu was 60.8% using 0.60 M HCl solution.

XRD patterns confirmed the formation of pure Sc oxide after calcination at 42 °C. The Dy oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. Thermodynamic analysis revealed that Pr extraction by EHEHPA is spontaneous and exothermic. Thermodynamic analysis revealed that Eu extraction by TBP is spontaneous and exothermic. The separation factor between Yb and Lu was 14.7 under optimized conditions. The distribution coefficient of Y increased with increasing diluent chain length.

Temperature significantly affected Y extraction, with optimum conditions at 61 °C. The stripping efficiency of Nd was 62.7% using 3.80 M HCl solution. An aqueous-to-organic ratio of 3:1 was found optimal for Gd loading. The Nd oxide nanoparticles were synthesized via adsorption and characterized by FTIR.

The Lu oxide nanoparticles were synthesized via ion exchange and characterized by FTIR. SEM images showed uniform Nd particle morphology with an average size of 187 nm.

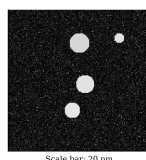


Figure 8. magnetic time stirring morphology samarium coating EHEHPA mass nanoparticles yield terbium thermodynamics phase holmium synergistic TBP.

The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. An aqueous-to-organic ratio of 1:2 was found optimal for Ce loading.

Solvent extraction of Dy from aqueous phase showed high efficiency at pH 4.4. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase. Temperature significantly affected Ho extraction, with optimum conditions at 37 °C.

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